MILITARY SPECIFICATION

CARTRIDGE, 20MM, DUMMY: M51A2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of dummy cartridge for use in 20mm, automatic, gun systems designed for ammunition using M103 type cartridge case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing, and Marking of Small Arms Ammunition
MIL-STD-651	- Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168	- Ammunition Lot Numbering
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC 1305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Frankford Arsenal, ATTN: SARFA-MDM, Philadelphia, Pennsylvania 19137 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T06:15Z
Check the source to verify that this is the currentversion before use.

2.1 (Cont)

DRAWINGS

ARMAMENT COMMAND

F7258941	- Method of Packing 100 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258942	- Method of Packing 200 Cartridges, 20mm, with M103 Type Case, in Bulk, in M548 Container
F7258968	- Method of Packing 100 Cartridges, 20mm, in M12 Type Links, in M548 Container
F7258969	- Method of Packing 100 Cartridges, 20mm, in M14 Type Links, in M548 Container
F7258970	- Method of Packing 100 Cartridges, 20mm, in M17 Type Links, in M548 Container
D7259079	- Marking on M548 Shipping and Storage Container for Cartridges, 20mm, Dummy
D11075758	- Cartridge, 20mm, Dummy, M51A2
LI 11075758	- Index of Inspection Equipment Lists for: Cartridge, 20mm, Dummy, M51A2

PUBLICATIONS

U.S. ARMY MATERIEL COMMAND

AMCR 715-505 - Ammunition Ballistic Acceptance Test Methods, Volume 8, Test Procedures for 20mm Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM B201-68 - Testing Chromate Coating on Zinc and Cadmium Surfaces

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 General. The cartridge shall be in accordance with Drawing (Dwg.) D11075758, referenced specifications, and other requirements specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:15Z
Check the source to verify that this is the current version before use.

3.2 Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (see 6.2)

3.3 First article. When specified, the contractor shall furnish sample cartridges for first article inspection and approval. (see 4.3 and 6.1)

3.4 Projectile extraction. The projectile shall not separate from the cartridge case at a tension load less than as specified on Dwg. D11075758.

3.5 Protective coating. The protective finish of the cartridge shall be as specified on Dwg. D11075758. The coating shall have a uniform appearance and be free from defects that may affect the protective value of the coating.

3.5.1 Corrosion resistance. The exterior coated surface of the assembly shall show no evidence of corrosion (red rust) after exposure to a 5 percent solution of atomized salt spray for 48 hours.

3.6 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Process defects. The cartridge shall be free of defects such as folds, wrinkles, buckles, bulges, deep scratches, cuts, dents, perforations, plating flaws, i.e., tarnish, scale, peeling, blisters, welding spatter and other forms of substandard workmanship.

3.6.2 Cleanliness. The cartridge and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:15Z
Check the source to verify that this is the current version before use.

4.3 First article inspection.

4.3.1 First article sample. Unless otherwise specified, a sample shall be delivered for inspection in accordance with the first article approval provisions of the contract and MIL-A-2550, and shall consist of 82 dummy cartridges. The dummy cartridges shall be identified in accordance with MIL-STD-1168.

4,3.1.1 Inspection, After inspection by the contractor and provisional acceptance by the Government activity, the sample will be inspected in accordance with the provisions of MIL-A-2550. Failure of the original first article sample to be in accordance with the applicable drawings and the requirements of Para. 3, as specified herein, will result in sample disapproval.

4.4 Quality conformance inspection.

4.4.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot. A lot shall consist of

a. Cartridges produced by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases, projectiles) having one manufacturer's symbol and one interfix number.

4.4.1.2 Lot identification. Each cartridge and each packed lot shall be identified in accordance with applicable drawings and MIL-STD-1168. Each packed lot shall be further identified by a National stock number assigned by the procuring activity.

4.4.2 Examination. One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge found to be defective shall be rejected.

4.4.2.1 Sampling and acceptance criteria. The sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235. To determine product acceptability, major or minor defects as listed herein may be considered collectively on a class basis or they may be considered individually. However, where major defects are considered collectively on a class basis, the acceptance number for any individual defect of the class shall be limited by the assigned Acceptable Quality Level (AQL) for individual defect associated with the class. (see 4.4.2.2.c) In addition, where MIL-STD-1235 is used sampling plans applicable to a class shall not be used for major defects. (see 6.3)

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:15Z
Check the source to verify that this is the current version before use.

4.4.2.2 AQL's. The following AQL's are assigned:

a. Individual defect characteristic basis:

Major	**0.40%**
Minor	**0.65%**

b. Defect characteristic, class basis:

Major	**1.50%**
Minor	**2.50%**

c. Individual defect characteristic associated with the class basis:

Major	**0.65%**

4.4.3 Classification of defects. 1/ The classification of defects shall be as specified herein.

4.4.3.1 Cartridge assembly (see Dwg. D11075758).

Categories and Defects	Method of Inspection	
Critical:		
1. Presence of live loaded cartridges(s)	**Visual**	
Major:		
101. Plating missing	**Visual**	
102. Less than minimum number of spot welds	**Visual**	
103. Overall length, max.	**Gage**	
104. Improper, incomplete, deformed or missing crimp (case-projectile)	**Visual**	2/
105. Profile and alignment, max.	**Gage**	3/
106. Corrosion	**Visual**	
107. Head configuration	**Visual**	
108. Weight	**Balance**	
109. Length, case head to centerline of crimp groove, max.	**Gage**	
110. Plug or cup missing (M103A1 case only)	**Visual**	
111. Stake missing, plug or cup (M103A1 case only)	**Visual**	
112. Depth, plug or cup, min. (M103A1 case only)	**Gage**	
Minor:		
201. Marking incorrect, incomplete, illegible or missing	**Visual**	
202. Protective coating damaged or incomplete	**Visual**	4/
203. Scratch, dent, buckle, bulge, wrinkle or fold in case	**Visual**	
204. Foreign matter, except corrosion	**Visual**	
205. Sharp edge, weld	**Visual/Tactile**	
206. Depth, plug or cup, max. (M103A1 case only)	**Gage**	

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:15Z
Check the source to verify that this is the current version before use.

4.4.3.1 (Cont)

1/ MIL-STD-651 shall apply in defining and evaluating visual defects as applicable.

2/ The crimp groove shall be free of raised sharp edges.

3/ A dead weight load of 60 pounds maximum may be used to insert the cartridge in the gage.

4/ The cartridge shall be classed defective if the basis metal is exposed on any portion of the cartridge.

4.4.4 Tests. The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in 4.5. Sample size and acceptance criteria for each test shall be as specified. Only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests.

Table I

Test		Sample Size Production Lot	Reqt Para
Projectile extraction	1/	50	3.4
Corrosion resistance	2/	32	3.5.1

1/ Projectile extraction. If two or more sample cartridges fail to comply with the minimum requirement the lot shall be rejected. If one cartridge of the sample fails to comply with the minimum requirement, a second sample consisting of the same number of cartridges as specified in the first sample shall be tested. The lot shall be rejected if two or more cartridges of the accumulated samples fail to comply with the minimum requirement. However, a cartridge case rupture at a tension less than the minimum required load will not be classified as a defective.

2/ If two or more units of the sample fail to comply with the corrosion resistance requirement, the lot shall be rejected, If one unit fails to comply with the requirement, a second sample consisting of the same number of units as specified for the first sample shall be tested. If in the accumulated samples, two or more units fail to comply with the corrosion resistance requirement, the lot shall be rejected.

4.4.5 Packaging inspection. The sampling and inspection for packaging, packing and container marking shall be in accordance with the requirements of MIL-STD-644, as applicable to the drawing.

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:15Z
Check the source to verify that this is the current version before use.

4.4.6 Inspection equipment. Index of Inspection Equipment List (LI) 11075758 identifies the applicable Inspection Equipment Lists (IEL) required to perform examination and tests prescribed herein. Equipment design(s) shall be in accordance with the applicable IEL code designations. The code designations are defined on Dwg. B11075228, a detail of LI 11075758. The provisions of MIL-A-2550 shall apply.

4.5 Test methods and Procedures.

4.5.1 Projectile extraction. The method of test shall be as prescribed in AMCR 715-505, Volume 8.

4.5.2 Corrosion resistance. The method of test shall be as prescribed in American Society for Testing and Materials (ASTM), ASTM B201, Testing Chromate Coating on Zinc and Cadmium Surfaces, Salt Spray Test, Procedure.

5. PACKAGING

5.1 Packing, Level A. The type of cartridge pack shall be in accordance with instructions from the procuring activity. Linked cartridges shall be packed as specified on Dwg. F7258968, F7258969 or F7258970. Bulk cartridges shall be packed as specified on Dwg. F7258941 or F7258942.

5.2 Marking. Marking on the packed container shall comply with Dwg. D7259079, as applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Title, number, and date of this specification.

6.1,2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements. (see 3.3 and 4.3)

6.1.4 Provisions for the examination, test and approval of first article sample.

6.1.5 Detailed packing and marking instructions. (see 5.1 and 5.2)

6.1.6 Provisions for the submission and approval of the manufacturing process. (see 3.2)

6.1.7 Provisions for the inclusion of MIL-STD-1167, Ammunition Data Cards on DD form 1423, Contract Data Requirement List.

6.2 Process deviation. A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

Source: https://assist.dla.mil -- Downloaded: 202610-22T06:15Z
Check the source to verify that this is the current version before use.

6.3 AQL's. The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.4 International agreements. Certain provisions of this specification are the subject of international standardization agreement STANAG 3585. When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes, with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Amy - MU
Air Force - 70

Preparing Activity:

Army - MU

Review Activities:

Army - MU
Air Force - 70

Project No. 1305-0805

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:15Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-46546C Cartridge, 20MM, Dummy: M51A2

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:15Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Phila., PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T06:15Z
Check the source to verify that this is the current version before use.